

# Dust Generator Bundschuh

For Dust Aerosol Generation during Inhalation Studies



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# Bundschuh Dust Generator

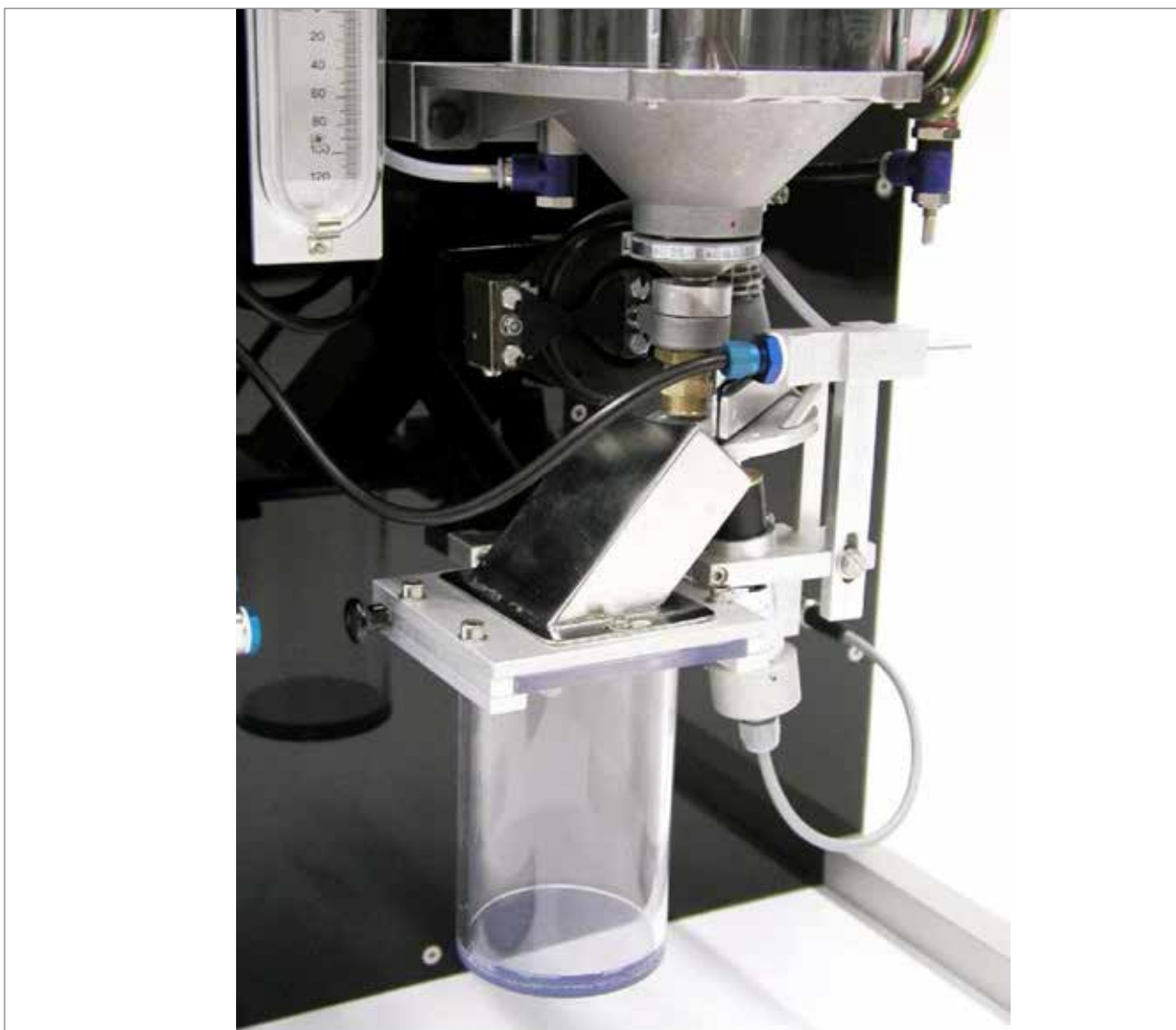
## General Information

The Bundschuh Dust Generator is suitable for free-flowing dusts and allows dispersion of up to 550 g/h. It has been designed to achieve a remarkably improved dosage both in the lower and the upper concentration range.

Basically the working principles of dust dosage and dispersion are used to produce dust aerosols suitable for inhalation experiments with Head Nose Only Exposure and Whole Body Exposure Systems.

The most important features are:

- a substance reservoir equipped with stirrer and ventilation for continuous dust conditioning during the experiment
- a perfect setting of aerosol concentration via both course and fine adjustment
- several versions of dosage plates – groove or holes – for different aerolization modes
- all components are mounted at a stable panel with support legs



Dosage components and recovery container of the dust generator

## Principle of Operation

Inside the reservoir the dust is continuously mixed by a mixer with variable speeds and ventilated using air. This ensures the most appropriate conditioning of the substance which should be aerosolized.

By means of a vibrating dosage unit the substance is forced out of the reservoir onto the dosage plate which is driven by a micro motor.

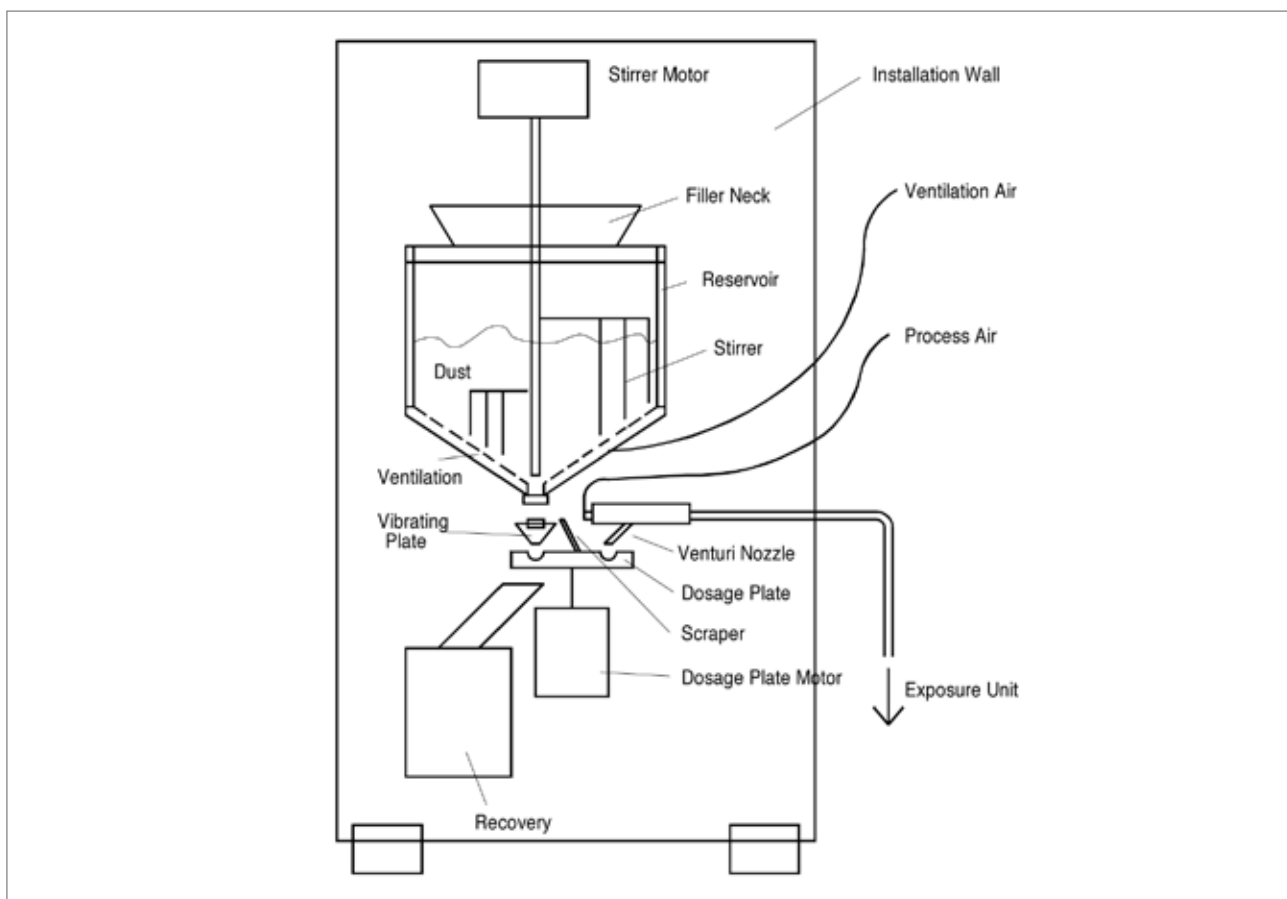
The dosage plate is equipped with holes or grooves along its periphery. The choice of the specific plate and the setting of the rotation speed enables an ideal aerosol dosage.

A scraper removes all the dust particles from the dosage plate which have not fallen into either the dosage holes or grooves.

These removed dust particles are collected in a recovery container. Particles which fall into the dosage holes or grooves are soaked up by a Venturi nozzle activated by the process flow.

The electronic process control unit allows the regulation of both the vibration intensity (coarse dosage) and the rotation speed of the dosage plate (one part of the fine dosage) via a feed-back loop to the optional system photometer in order to adjust the dosage of the specific test substance.

The dust generator has been designed to disperse amounts of 0.1 to 550 g per hour and to produce substance concentrations between 0.001 and 3 g per m<sup>3</sup> process air.



Scheme of Bundschuh Dust Generator

Due to the varying physical properties of the substances concerned, the user has to test and ascertain substance specific system settings during pre-tests.

The preparation of the substance prior to aerosolization also plays an important role. If the substance particles are not small enough to be delivered into the lungs via respiration (MMAD < 3 µm) it is recommended to improve the particle size distribution before the experiment by an appropriate pre-separation procedure for larger particles (e.g. via a jet mill). Please ask us for assistance in this matter.

In some cases the usage of a carrier dust is favourable meaning that the "active substance" is mixed with another dust, e.g. lactose. In experiments according to this design both particle distributions have to be monitored.

Key attention must be paid to the MMAD of the active substance in order to reach the alveoles of the animals.



Reservoir with stirrer



Stirrer and ventilation – porous yellow stripes



Dosage plate (holes) with scraper and Venturi nozzle

## Ordering Information

| Cat. No.     | Description                                                                                    |
|--------------|------------------------------------------------------------------------------------------------|
| 700700-SG-01 | <b>Bundschuh Dust Generator</b><br>For dust aerosol generation up to 3,000 mg/m <sup>3</sup> . |

## Partial List of Users

- Almirall Prodesfarma, S.A., St. Feliu de Llobregat - Barcelona, Spain
- Astra Zeneca R&D Charnwood, Loughborough, Leicestershire, Great Britain
- Bioagri Pharma, Planaltina, DF, Brazil
- Boehringer Ingelheim Pharma GmbH & Co. KG, Biberach, Germany
- Charles River Laboratories, Tranent, Edinburgh, Great Britain
- Chiesi Farmaceutici S.p.A., Parma, Italy
- CIEMAT, Madrid, Spain
- Dow Corning Corporation, Midland, MI, USA
- DSO National Laboratories, Singapore, Singapore
- Elbion AG, Radebeul / Dresden, Germany
- GSF-Forschungszentrum für Umwelt & Gesundheit, GmbH, Neuherberg, Germany
- IVAX Drug Research Institute Ltd., Budapest, Hungary
- KT&G Central Research Institute, Yuseong-gu, Daejeon, South Korea
- LAB International Research Center Hungary Ltd., Veszprem, Hungary
- Lovelace Respiratory Research Institute, Albuquerque, NM, USA
- Mahidol University, Bangkok, Thailand
- Pulmatrix Inc., Lexington, MA, USA
- Schering-Plough Research Institute, Kenilworth, NJ, USA
- The Dow Chemical Company, Midland, MI, USA
- UFZ, Leipzig, Germany
- Universidad de Salamanca, Salamanca, Spain
- University of California, Irvine, CA, USA
- University of Minnesota, Minneapolis, MN, USA



**TSE Systems offers additional modules to create a complete exposure system. Please find out more about:**

■ **Head Nose Only Exposure Units**

Modular, stackable structure for different size conditioning units. Available for animal sizes from mouse to dog with directed-flow principle

■ **Whole Body Exposure Units**

Robust, user-friendly design for homogeneous test substance distribution. Different volumes available for variable animal sizes and test animal numbers

■ **Cell Culture Exposure System**

Ideally suited system for screening tests with tissue, using Transwell® inserts of varying sizes. Extended viability of cultures by heating/humidification

■ **Nanoparticle Analysis**

TSE System offers instrumentation to analyze nanoparticle size and concentration, such as Condensation Particle Counter or Differential Mobility Analyzer

■ **Daco Inhalation Software**

Control and recording software for inhalation systems with multi-control mode for system extension. Advanced safety and GLP compliance features

■ **Sensors and Controllers**

TSE Systems supplies a number of environmental sensors and mass flow controllers specifically designed for each inhalation system

*Specifications subject to change without notice*

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## Service & Warranty

TSE Systems offers a Two (2) Years ALL-IN Premium Warranty with all new products, including:

- 24/7 technical hotline
- Remote maintenance and update function
- On-site visits upon necessity
- Free replacement parts

After the expiry of the warranty period, TSE Systems offers comprehensive extensions of the warranty or economical maintenance and repair contracts to ensure the continued smooth running of your instruments. Please contact us for further details.

